MAE 4736
9/14/18
Lecture 9

For <u>any</u> system

AMB true, trust it.
• Any point C

$$\sum \vec{M}^{ext}_{/c} = \begin{cases} \sum \vec{r}_{i/c} \times m_i \vec{a}_i & \text{discrete} \\ \int \vec{r}_{/c} \times \vec{a}\, dm & \text{continuous} \end{cases}$$

to the extent that mechanics is true

$$\dot{\vec{H}}_{/c} \stackrel{?}{=} \frac{d}{dt}(\vec{H}_c)$$

Good definition of $\vec{H}_{/c}$

$$\vec{H}_{/c} = \sum \vec{r}_{i/c} \times m \vec{v}_{i/c}$$

$\vec{r}_{i/c} = \underbrace{\vec{r}_i - \vec{r}_c}_{\vec{r}_{i/o} - \vec{r}_{c/o}}$

$\vec{v}_{i/c} = \underbrace{\vec{v}_i - \vec{v}_c}_{\vec{v}_{i/\mathcal{F}} - \vec{v}_{c/\mathcal{F}}}$

<u>Good for:</u>

1) C is a fixed point in $\mathcal{F}$
2) $\underbrace{\text{C is G = CoM}}_{\text{can be accelerating}}$
3) C has const. vel in $\mathcal{F}$
4) C has $\vec{a}$ccell $\parallel \vec{r}_G - \vec{r}_C$

<u>2 Factoids:</u>

$$\vec{H}_{/c} = \vec{H}_{G/c} + \vec{H}_{/G}$$

$$\dot{\vec{H}}_{/c} = \dot{\vec{H}}_{G/c} + \dot{\vec{H}}_{/G}$$

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